

Shane Thakkar

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SUMMARY

Analytics & AI professional with 2.5 years of hands-on experience in Python, SQL, and machine learning across sports, finance, and telecom. Built automations saving 10+ hours per week, delivered ML models reaching 90% accuracy, and developed venue clustering that guided multimillion-dollar LA28 investments. Independently ship statistically rigorous full-stack data products, including a public NFL player grading platform with 20,000+ player-season grades, Bayesian shrinkage, and audit-defensible methodology.

EDUCATION

B.S. Business Analytics and Artificial Intelligence – Data Science Track

University of Texas at Dallas • Richardson, TX • May 2026

SKILLS

Programming & Analytics: Python, Pandas, NumPy, SciPy, TypeScript, SQL, R, Git, Bash/UNIX, Jupyter

LLM & AI Workflow: Claude Code, Cursor, LangChain, Prompt Engineering, RAG, Multi-Agent Systems

Machine Learning: Scikit-learn, TensorFlow, Keras, PyTorch, PyMC, XGBoost, Optuna

Statistical Methods: Bayesian Inference, Time-Series Forecasting, Classification, Clustering, Regression Analysis

Big Data & Cloud: Databricks, Snowflake, Spark, Azure, SQL Server, Docker, Hadoop, Hive

Visualization & Tools: Tableau, Power BI, Matplotlib, Plotly, Excel, Stata, Alteryx, Power Automate

EXPERIENCE

Business Intelligence Intern

Legends Global

July 2025 - December 2025, Frisco, TX

- Analyzed 1M+ POS records across 8+ professional sports venues using Python and Pandas, cleaning and standardizing data to perform category-level performance analysis that identified high and low-performing products and drove inventory, pricing, and seasonal planning decisions.
- Applied Market Basket Analysis and Pareto principles in Python to evaluate FC Dallas concessions, discovering product pairings for bundling strategies and identifying 30% of SKUs as tail products generating <15% of revenue.
- Consolidated 2 years of POS data from 3 systems for Del Mar Racetrack, building reports and SWOT analyses that identified key revenue drivers and performance trends, presenting findings to stakeholders to inform strategy.
- Modeled F&B performance for ~15 LA28 venues in Python by normalizing historical per-cap and attendance data, then clustering venues into priority tiers that informed executive investment and risk decisions.
- Automated weekly SQL Server reconciliation between game-day flash reports and POS transaction data across venues, flagging discrepancies in audit logs and eliminating ~10 hours per week of manual validation work.
- Pioneered transparent reporting practices by adding methodology sections to all analyses, documenting data sources, assumptions, and formulas for stakeholder validation; later adopted as the team standard.

Project Management and Data Analytics Intern

Supreme Lending

June 2024 - November 2024, Dallas, TX

- Built Excel-based project management system coordinating 100+ test cases across departments during loan origination software upgrade, tracking completion KPIs through automated dashboards ensuring defect-free rollout.
- Built Power Automate workflow to automate Azure DevOps user story creation, reducing manual request intake time and enabling automated stakeholder tracking and notifications.
- Led team of 14 interns in 30-minute executive pitch proposing university mortgage lending course, presenting Excel pivot-table analysis of 300+ student surveys showing ~60% interest.

Data Integration and Analytics Intern

TruGen

May 2023 - May 2024, Frisco, TX

- Cleaned and structured wireless network performance data across 10+ regions using SQL, preparing datasets for machine learning models that achieved 90% classification accuracy in signal quality prediction.

RELEVANT PROJECTS

NFL Player Grading Platform

nfl-grades.shanethakkar.com

Built and deployed a full-stack NFL analytics data product generating 0–100 player grades by position and season across 12 positions and 2016–2025 data, producing 20,000+ player-season grades.

- Engineered a six-stage Python grading pipeline using pandas, SciPy, and SQLAlchemy, applying volume-weighted empirical Bayes shrinkage, qualified-cohort z-score normalization, magnitude-normalized composites, sigmoid calibration, and Kalman smoothing to produce stable, comparable grades.
- Evaluated 190+ candidate metrics against four criteria: year-over-year reliability, cross-sectional discrimination, independence, and predictive validity vs. Pro Bowl selections; shipped 52 components to production at a 27% acceptance rate with rejections logged for audit defensibility.
- Deployed a Next.js 15, React 19, and TypeScript web app on Vercel with Neon serverless Postgres, 25+ idempotent SQL migrations, mobile-responsive sortable leaderboards, 647 AI-generated player headshots via the Replicate API, and composite-indexed queries returning in under 50ms.

EdgarRisk: Corporate Distress Detector

shanethakkar.com/articles/edgar-risk

Built a Python NLP pipeline detecting corporate distress from SEC 10-K filings on the EDGAR system, validated across 24 Chapter 11 bankruptcies and 42 sector-matched survivors spanning 15 industries.

- Engineered a 4,300-line Python NLP pipeline across 21 scripts detecting corporate distress from SEC 10-K Risk Factor disclosures via a three-signal composite combining TF-IDF cosine novelty scoring, Loughran-McDonald sentiment analysis, and peer-relative percentile-rank scoring against sector-matched cohorts.
- Validated against 24 corporate failures and 42 sector-matched survivors across 15 industries with 79% recall, precision climbing from 46% to 69% at 2–3 year horizons, a held-out test on Spirit Airlines, and a 78-pair false-positive analysis.
- Demonstrated 1–3 year leading-indicator value via longitudinal follow-up: 27% of in-sample false positives subsequently underwent material distress events, correctly flagging Nordstrom, Walgreens, and CVS years before their 2024–2025 take-privates and stock crashes.

NFL Fourth-Down Decision Analysis

shanethakkar.com/articles/fourth-down

Analyzed 107,000+ NFL fourth-down decisions from 1999–2025 using Win Probability Added to grade 167 head coaches on decision quality, with three deployed interactive web apps and a published article.

- Defined original Decision Quality Score and Optimal Decision Rate metrics on a WPA-based framework, binning 107,000+ plays across a four-dimensional state lattice of field position, yards-to-go, score differential, and time remaining with exponential recency weighting and minimum-sample filtering.
- Trained three XGBoost regressors predicting expected WPA for go-for-it, punt, and field-goal decisions across yardline, distance, score, and time-remaining features on 1999–2023 data with 2024–2025 held out for validation.
- Deployed three interactive Streamlit applications: a decision calculator that scores any game-state input, an XGBoost-powered decision boundary heatmap visualizing optimal calls across the field, and a sortable explorer ranking 167 head coaches by Decision Quality Score and Optimal Decision Rate.

F1 Bayesian Driver Rankings

shanethakkar.com/articles/f1

Built a hierarchical Bayesian model decomposing driver skill from car performance across 12 seasons of Formula 1 race data from 2014–2025, covering 46 drivers and 123 team-seasons.

- Engineered the model in Python using PyMC and FastF1, applying a Student-T likelihood for heavy tails, partial pooling hyperpriors on driver and team-season variance, and a DNF-fault dummy to prevent penalizing drivers for mechanical failures outside their control.
- Validated convergence with $\hat{R} \approx 1.0$ across NUTS chains, temporal holdout RMSE of 3.114 vs. baseline 3.146, and posterior predictive checks reproducing the empirical residual distribution.
- Surfaced findings via three native React and SVG interactive charts in the published article: Hamilton ranking #1 with 85% posterior confidence, era-specific constructor dominance across 123 team-seasons, and the "Verstappen Paradox" where the metric rewards on-track recovery over qualifying pace.

CERTIFICATIONS

[IBM AI Engineering Professional Certificate](#) · [Snowflake Hands-On Essentials: Data Warehousing Workshop](#) · [Professional Scrum Master™ I](#)